

# *Week 4*

Keel Bone Chicken Project

06/08/26



College of Engineering

# ***This Week***

## Weekly Progress

- Data Collection Updates
- Database Updates
  - Database Setup
  - Database Extraction
  - Database Beta Version
- Sensor Testing Updates
- Next Week

# *Data Collection Updates*

## Agriculture Updates on Farm

- Device taken to farm
  - Undergraduate student allowed to test and collect data
  - Collected for around ~2 hours
- Documentation for device given to Sara (Post-doc) and Undergraduate
  - Hopefully setup was successful
  - May be bugs
- Sara requested I check the data collected

# Database Updates

## Database Setup

- Connecting to MongoDB compass in the new cluster
- Added secure user
  - User will be able to change databases
  - Others will only be able to read databases

### Connecting with MongoDB Compass

I don't have MongoDB Compass installed

I have MongoDB Co

#### 1. Select your operating system and download MongoDB Compa

Windows 64-bit (10+) ▼

Download Compass (1.49.8) or Copy download URL

Compass is an interactive tool for querying, optimizing, and analyzing your Mon

#### 2. Copy the connection string, then open MongoDB Compass

Edit User: huangnathan69\_db\_user@admin

Update a database user to grant an application or user access to databases and collections in your clusters in this Atlas project. Granular access control can be configured with default privileges or custom roles. You can grant access to an Atlas project or organization using the corresponding [Access Manager](#)

#### Authentication Method

Password

Certificate

AWS IAM

Federated Auth  
(MongoDB 7.0 and up)

MongoDB uses [SCRAM](#) as its default authentication method.

#### Password Authentication

huangnathan69\_db\_user

\*\*\*\*\*

SHOW

Autogenerate Secure Password

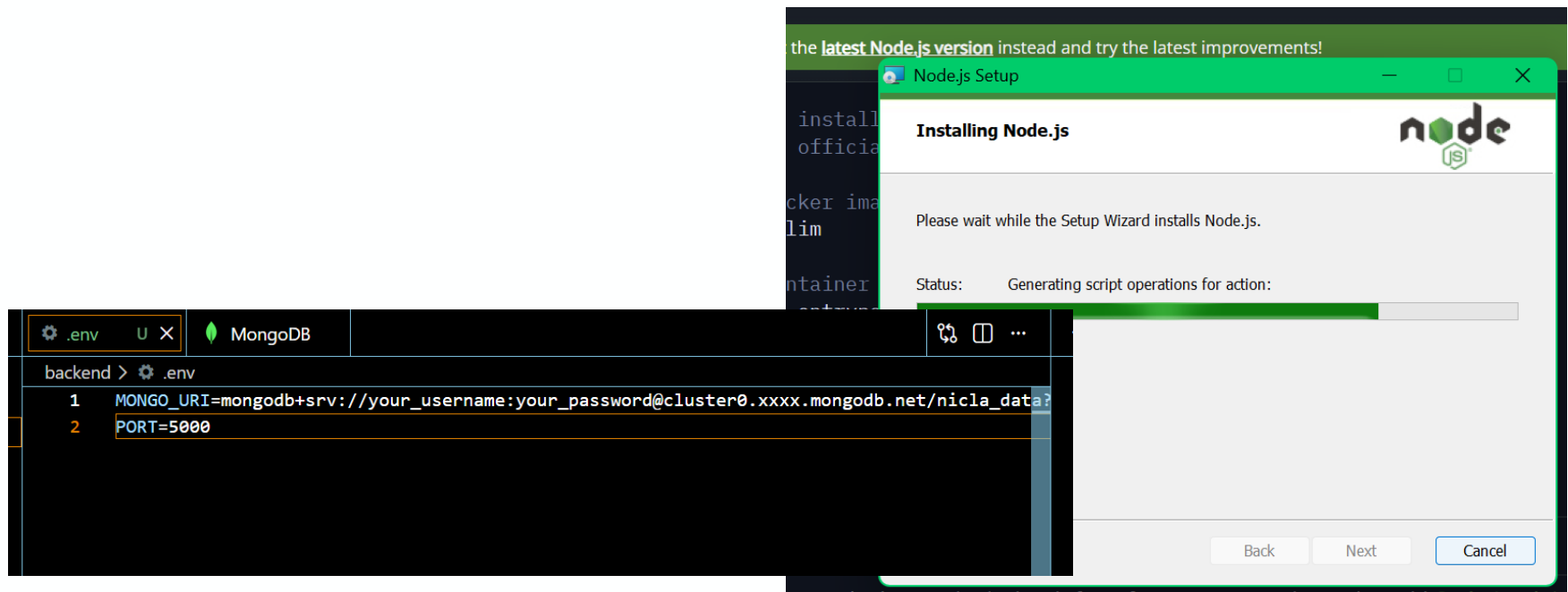
Copy

Help Description

# Database Updates

## Copied Signing Key to Database

- Set the signing key in Visual Studio Code
- Copied into the .ENV file
- Downloaded Node.js for development



# Database Updates

## Database Extraction

- Began calling attributes in the terminal to be transferred over
- Database calls appear to be working
- Have to write unit tests to confirm this

```
    gyroscope: { x: Number, y: Number, z: Number }
  });

const SensorData = mongoose.model('SensorData', SensorDataSchema);

// API Route to receive data from Frontend
app.post('/api/sensor-data', async (req, res) => {
  try {
    const newData = new SensorData(req.body);
    await newData.save();
    res.status(201).json({ success: true, message: "Data logged successfully!" });
  } catch (error) {
    res.status(500).json({ success: false, error: error.message });
  }
});

// API Route to verify data (Testing)
app.get('/api/sensor-data', async (req, res) => {
```

```
\natha\School\GRA_Polytechnic\Biomedical-Device-Database>npm init -y
C:\Users\natha\School\GRA_Polytechnic\Biomedical-Device-Database\package.json:

  "name": "biomedical-device-database",
  "version": "1.0.0",
  "description": "Database Setup (MongoDB)",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "Nathan",
  "license": "MIT",
  "repository": {
    "type": "git",
    "url": "https://github.com/nathan-schoen/biomedical-device-database.git"
  }
}
```

# Database Updates

## Database Beta Version

- Will host a localhost version on server 5000
- Set the backend\_url to be constant for testing
- Only applicable to my development environment

```
let currentSensorPayload = {  
  temperature: null,  
  humidity: null,  
  pressure: null,  
  gas: null,  
  accelerometer: { x: 0, y: 0, z: 0 },  
  gyroscope: { x: 0, y: 0, z: 0 }  
};  
  
// Target backend endpoint  
const BACKEND_URL = 'http://localhost:5000/api/sensor-data';
```

# *Sensor Testing Updates*

## Inquiries on force testing machine

- Force in-out testing, not sufficient equipment at lab
- More precise equipment at Prof. Wenzhuo Wu's lab
- Will contact him later this week about a visit



# *Next Week*

## Next Week's Plan

- Analyze data collected from web logger in the Agriculture trial
  - See if any bugs arose
  - Verify the entire length of data is being collected
- Contact Professor Wu about the Force sensor testing unit

# *Thank You*

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